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Basic needs: locomotion, power and navigation

A nanorobot is essentially a machine at the nanometre or molecular scale composed of nano-scale components, capable of actuation, sensing, signalling, information processing, intelligence and swarm behaviour. So, like all robots, nanorobots also need systems for locomotion, navigation and power. Here is how these functions are likely to be handled in nanorobots, especially those injected into patients for some form of treatment:

Navigation

- External systems or onboard systems may be used for sensing and controlling their movement.
- Ultrasonic signals may be used to detect the nanorobot's location and direct it to the right destination.
- Engineers at the Ecole Polytechnique de Montreal have showed how to detect, track, control, power and propel nanorobots using magnetic resonance imaging (MRI) devices. Because medicine is one of the main applications of nanorobots, and most hospitals have MRI machines, this might become the industry standard.
- Doctors might also track nanorobots by making these emit a radioactive dye as they travel into the patient's bloodstream. A fluoroscope could be used to detect the dye.
- X-rays, radio-waves, microwaves or heat may also be used to detect nanorobot movement.
- Internal sensors, such as chemical or spectroscopic sensors, might help the nanorobot detect certain aspects of its environment and thereby judge its path.
- Well into the future, nanorobots might include a miniature camera. The images captured by this camera would help the operator to trace and control the nanorobot's movement remotely.

Power

- Onboard or external systems might be used.
- Nanorobots could draw power from the patient itself—body heat or movement could help charge them.
- A nanorobot with mounted electrodes could use electrolytes in the patient's blood to work like a battery.
- Other chemical reactions with blood could be used to produce energy.
- Engineers are also working on building smaller capacitors to power nanorobots.
- In the rare possibility that a nanorobot is connected to a control system through a wire, power could also be supplied through the same wire.
- Light can also be used as an external power source.
- Microwaves, ultrasonic signals or magnetic fields could also be converted into power for the nanorobot.
- A nanorobot with a piezoelectric membrane could pick up ultrasonic signals and convert them into electricity.

Locomotion

- Since a nanorobot injected into a human body has to usually travel against the flow of blood, it needs a strong propulsion system.
- The locomotion must also be safe—even a slight detour or improper speed can cause harm to the patient.
- Scientists are watching organisms like Paramecium to understand how they move, so that nanorobots can be modelled on nature-inspired techniques.
- One team of scientists has developed a micro-robot that uses small appendages to grip and crawl through blood vessels. The scientists manipulate the arms by creating magnetic fields outside the patient's body.
- Miniaturised jet pumps could use blood plasma to push the nanorobot forward. Another option is to use an electromagnetic pump for this purpose.
- Nanorobots could move around by using a vibrating membrane. By alternately tightening and relaxing tension on the membrane, a nanorobot could generate small amounts of thrust for it to move.

communication, navigation, manipulation, locomotion and onboard computation, have been presented only in the medical context so far," says Rajeev Karwal, CEO and founder, Milagrow Business & Knowledge Solutions.

We have to wait and see how the nano-bug spreads to other fields. "The

world is just now waking up to the potential of this technology. Now factors like funding, standards and control will play a role in speeding up the development and adoption, and bringing down the costs," Karwal adds. ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai